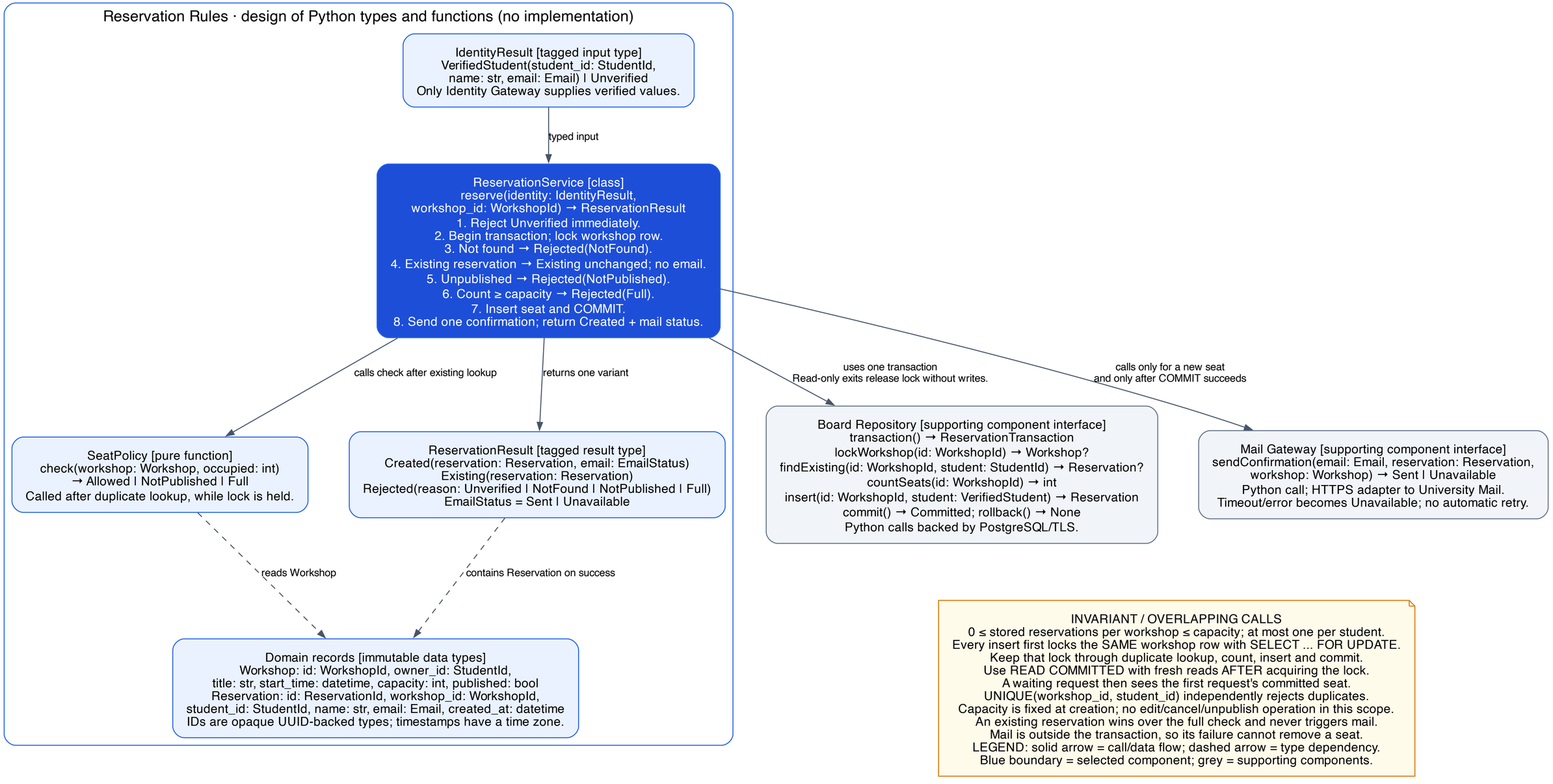
04 · Code sketch
Parent: Reservation Rules [Component inside Board API]
Ali Alyaarbi · 25011226



INVARIANT / OVERLAPPING CALLS
0 ≤ stored reservations per workshop ≤ capacity; at most one per student.
Every insert first locks the SAME workshop row with SELECT ... FOR UPDATE.
Keep that lock through duplicate lookup, count, insert and commit.
Use READ COMMITTED with fresh reads AFTER acquiring the lock.
A waiting request then sees the first request's committed seat.
UNIQUE(workshop_id, student_id) independently rejects duplicates.
Capacity is fixed at creation; no edit/cancel/unpublish operation in this scope.
An existing reservation wins over the full check and never triggers mail.
Mail is outside the transaction, so its failure cannot remove a seat.
LEGEND: solid arrow = call/data flow; dashed arrow = type dependency.
Blue boundary = selected component; grey = supporting components.